CStats - Lab #5

March 24, 2026

Exercise 1.
In order to test whether four brands of gasoline give equal performance in terms of mileage, each of three cars was driven with each of the four brands of gasoline. Then each of the (3)(4) = 12 possible combinations was repeated four times. The number of miles per gallon for each of the four repetitions in each cell is recorded in Table 1. Test the hypotheses H_{AB}: no interaction, H_A: no row effect, and H_B: no column effect, each at the 5% significance level.

	Brand 1	**Brand 2**	**Brand 3**	**Brand 4**
Car 1	31.0, 24.9, 26.2, 28.8	26.3, 30.0, 25.2, 31.6	25.8, 29.4, 24.5, 24.8	27.8, 27.3, 28.2, 30.4
Car 2	30.6, 29.5, 30.8, 28.9	25.5, 26.8, 27.4, 29.4	26.6, 23.7, 28.2, 26.1	28.1, 27.1, 31.5, 29.1
Car 3	24.2, 23.1, 26.8, 27.4	27.4, 28.1, 26.4, 26.9	25.2, 26.7, 27.7, 28.1	26.3, 26.4, 27.9, 28.8

Table 1: Gasoline Performance by Car and Brand

Exercise 2.
The output in Table 2 was obtained from a computer program that performed a two-factor ANOVA on a factorial experiment.
(a) Fill in the blanks in the ANOVA table. You can use bounds on the p-values.
(b) How many levels were used for factor B?
(c) How many replicates of the experiment were performed?
(d) What conclusions would you draw about this experiment?

Exercise 3.
Show that the unbiased estimator of the variance σ^2 from a sample of size $n = 2$ is one half of the square of the difference of the two observations. Thus, show that, if a 2^k design is replicated, say, with X_{i1} and X_{i2} , $i = 1, 2, ..., 2^k$, then the estimate of the common σ^2 is

$$\frac{1}{2^{k+1}} \sum_{i=1}^{2^k} (X_{i1} - X_{i2})^2 = \text{MSE}$$

Two-way ANOVA: y **versus** A, B

Source	DF	SS	MS	F	p
A	1	?	0.0002	?	?
B	?	180.378	?	?	?
Interaction	3	8.479	?	?	0.932
Error	8	158.797	?		
Total	15	347.653			

Table 2: Two-way ANOVA Table

where MSE refers to mean squared error. Under the usual assumptions, this equation implies that each of $2^k[\mathrm{A}]^2/\mathrm{MSE}$, $2^k[\mathrm{B}]^2/\mathrm{MSE}$, $2^k[\mathrm{AB}]^2/\mathrm{MSE}$, and so on has an $F(1, 2^k)$ distribution under the null hypothesis. This approach, of course, would provide tests for the significance of the various effects, including interactions.

Exercise 4.

The experiment shown in Table 3 is a variation of the study on polysilicon doping. The response variable is base current. Is there evidence (with 0.05) indicating that either polysilicon doping level or anneal temperature affects base current? Finally, define a model for this experiment and conclude your results.

Polysilicon Doping (ions)	Anneal Temperature (°C)		
	900	950	1000
1×10^{20}	4.60	10.15	11.01
	4.40	10.20	10.58
2×10^{20}	3.20	9.38	10.81
	3.50	10.02	10.60

Table 3: Data - Polysilicon doping vs Anneal Temperature